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AUTONOMOUS INSPECTION OF A SURFACE TOPOLOGY OF AN AIRFOIL OF A GAS TURBINE ENGINE

Abstract

Apparatus and associated methods relate to autonomous devices configured to inspect surface topologies of airfoils of a gas turbine engine. The autonomous device moves across the airfoil while remaining coupled thereto while sensing the surface topology of the airfoil using a plurality of topological sensors. Each of the plurality of topological sensors includes a piezoelectric sensor that generates an electrical signal indicative of a loading to the piezoelectric sensor, as well as an airfoil contacting member. The airfoil contacting member extends between the piezoelectric sensor and the airfoil when the body is coupled thereto. The airfoil contacting member provides changes to the loading to the piezoelectric sensor in response to changes in the surface topology of the airfoil as the body moves thereacross.

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Background/Summary

BACKGROUND

[0001] Gas turbine engines are used in many, if not most, commercial aircraft as the source of propulsion. Such engines have many and various sets of airfoils, which are used to perform various functions. Many of these airfoils are grouped into disks (or stages) of airfoils that rotate at high rotational rates about an axis of these gas turbine engines. For example, compressor blades are rotating airfoils that are principally used to compress air before air enters a combustion chamber. Turbine blades are rotating airfoils that extract mechanical energy from the expanding gases of the combustion process to turn other blades, such as, for example, compressor blades and fan blades. Turbofan engines are gas turbine engines equipped with fan blades, which are rotating airfoils that provide much, if not most, of the propulsion generated by such turbofan engines. Many gas turbine engines also have non-rotating blades that are group into stages of airfoils as well. Such non-rotating blades, called stator blades, are non-rotating airfoils that control rotational direction of the gases within gas turbine engines so as to prevent stalling of the rotating airfoils and to increase efficiency of the gas turbine engines. Such non-rotating stages are often interposed between rotating stages.

[0002] During operation of gas turbine engines, these various airfoils can be worn and or damaged. Ingesting dust, ash, or larger foreign objects can result in abrasion, pitting, or destruction of these airfoils. Because the integrity of these airfoils is important for safe and effective operation of gas turbine engines, routine inspections thereof are performed. Some such inspections involve an inspector utilizing a borescope to assist visualization of these engine parts. When possible, the inspector might even crawl into the engine to visually inspect the engine and/or to drag his/her finger across each blade, in an attempt to feel for any imperfections. Calibrating the accuracy of such manual inspections can be difficult. In some examples, inspections are performed by cameras, which involves taking high resolution imagery, which can supplement human visual evaluation by performing computational image processing so as to be able to detect tiny defects that might be present. Such inspections are typically only performed during scheduled maintenance operations, due to the time required and equipment needed for thorough inspection.

SUMMARY

[0003] Some embodiments relate to an autonomous device for inspecting surface topology of an airfoil of a gas turbine engine. The autonomous device includes a body, means for coupling the body to the airfoil in a manner that permits movement of the body across the airfoil while remaining coupled thereto, and means for moving the body across the airfoil while remaining coupled thereto. The autonomous device also includes a plurality of topological sensors extending from the body to the airfoil when the body is coupled thereto. Each of the plurality of topological sensors includes a piezoelectric sensor that generates an electrical signal indicative of a loading to the piezoelectric sensor. Each of the plurality of topological sensors also includes an airfoil contacting member extending between the piezoelectric sensor and the airfoil when the body is coupled thereto, thereby providing changes to the loading to the piezoelectric sensor in response to changes in the surface topology of the airfoil as the body moves thereacross. The autonomous device also includes a surface topology calculator attached to the body and electrically connected to the plurality of piezoelectric sensors and configured to locate defects in the surface topology of the airfoil based on the electrical signals generated by the plurality of piezoelectric sensors.

[0004] Some embodiments relate to a method for inspecting surface topology of an airfoil of a gas turbine engine. The method includes adhering a base of an autonomous device to the airfoil in a manner that permits movement of the body across the airfoil while remaining coupled thereto. The method continues by moving the base of the autonomous device across the airfoil while remaining coupled thereto. The method includes extending a plurality of topological sensors from the body to the airfoil when the body is coupled thereto. For each of the plurality of topological sensors, the method includes providing changes to a loading to a piezoelectric sensor in response to changes in

the surface topology of the airfoil as the body moves thereacross. For each of the plurality of topological sensors, the method also includes generating, via the piezoelectric sensor, an electrical signal indicative of a loading to the piezoelectric sensor. The method includes locating, via a surface topology calculator, defects in the surface topology of the airfoil based on the electrical signals generated by the plurality of piezoelectric sensors.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0005] The material described herein is illustrated by way of example and not by way of limitation in the accompanying figures. For simplicity and clarity of illustration, elements illustrated in the figures are not necessarily drawn to scale. For example, the dimensions of some elements may be exaggerated relative to other elements for clarity. Further, where considered appropriate, reference labels have been repeated among the figures to indicate corresponding or analogous elements. In the figures:

[0006] FIG. **1** is a perspective view of a swarm of autonomous devices inspecting surface topologies of fan blades of a gas turbofan engine.

[0007] FIG. **2** is a close-up or magnified view of the surface topology of a damaged fan blade of a gas turbofan engine.

[0008] FIGS. **3**A and **3**B are a perspective view and a block diagram, respectively, of an embodiment of an autonomous device configured to inspect the surface topology of an airfoil of a gas turning engine.

[0009] FIG. **4** is a schematic diagram of an embodiment of a topological sensor, which is configured for static operation, used to sense a surface topology of an airfoil.

[0010] FIG. **5** is a schematic diagram of an embodiment of a topological sensor, which is configured for dynamic operation, used to sense surface topology of an airfoil.

[0011] FIG. **6** is a graphical example of topology data of an airfoil that is obtained by a single robot.

DETAILED DESCRIPTION

[0012] Inspecting airfoils of gas turbine engines more frequently than is permitted by regularly scheduled maintenance sessions would increase safety and longevity of such gas turbine engines. Such frequent inspections could be performed if such inspections could be quickly and easily performed while an aircraft is parked at a gate of an airport, for example. Such quick and easy inspections could be made possible if performed by small devices that could autonomously navigate across an airfoil of a gas turbine engine and quickly survey such an airfoil. By deploying a swarm of such small autonomous devices, many airfoils could be simultaneously surveyed. By using algorithms that do not require powerful processing engines, such small autonomous devices could possess processors capable of performing such algorithmic operations.

[0013] Apparatus and associated methods described herein relate to such autonomous devices configured to inspect surface topologies of airfoils of a gas turbine engine. The autonomous device moves across the airfoil while remaining coupled thereto while sensing the surface topology of the airfoil using a plurality of topological sensors. Each of the plurality of topological sensors includes a piezoelectric sensor that generates an electrical signal indicative of a loading to the piezoelectric sensor, as well as an airfoil contacting member. The airfoil contacting member extends between the piezoelectric sensor and the airfoil when the body is coupled thereto. The airfoil contacting member provides changes to the loading to the piezoelectric sensor in response to changes in the surface topology of the airfoil as the body moves thereacross.

[0014] FIG. **1** is a perspective view of a swarm of autonomous devices inspecting surface topologies of fan blades of a gas turbofan engine. In FIG. **1**, gas turbofan engine **10** includes spinner **12**, fan **14** and nacelle **16**. Fan **14** includes a plurality of fan blades **18**. Each of the swarm of autonomous devices **20** is coupled to a fan blade **18** in a manner that permits movement of autonomous device **20** across fan blade **18** while remaining coupled thereto. Autonomous devices **20** are configured to inspect surface topologies of the fan blades **18** to which autonomous devices **20** are coupled. To inspect the surface topologies, each autonomous device **20** has topological sensors that generate electrical signals indicative of the surface topology of fan blade **18** as autonomous device **20** traverses fan blade **18**.

[0015] In the embodiment depicted in FIG. **1**, each autonomous device **20** has legs **22** that move autonomous device **20** across fan blade **18**. Legs **22** are configured to both couple autonomous device **20** to fan blade **18** and to move autonomous device **20** across fan blade **18**. In some embodiments, legs **22** have suction cups that couple each autonomous device **20** to fan blade **18** using evacuated chambers with such suction cups. In other embodiments, legs **22** affix to fan blade **18** via an adhesive and/or via a contacting surface that couples using van der Waals forces. In the depicted embodiment, legs **22** can be articulated so as to push and/or pull autonomous device **20** across fan blade **18**, and then be repositioned to a new location upon fan blades **18**. Such alternative pushing/pulling and repositioning can be coordinated with coupling/uncoupling to fan blades **18**. For example, miniature vacuum pumps (possibly manufactured by nano manufacturing techniques, using piezoelectric crystal vibrators, for example) can provide suction coupling of legs **22** to fan blade **18** while legs **22** are pushing and/or pulling autonomous device **20** across fan blade **18**. Then, air can be provided to such suction cups to relieve suction while legs **22** are repositioned to new locations on fan blade **18**. Such coordination of coupling and repositioning can be performed in a manner such that at least four of the six legs **22**, as depicted in the FIG. **1** embodiment, are at all times providing coupling of autonomous device **20** to fan blade **18**. Other means for moving autonomous device **20** across airfoil **18**, such as fan blades **18**, will be described below, with reference to FIGS. **3**A and **3**B.

[0016] FIG. **2** is a close-up or magnified view of the surface topology of a damaged fan blade of a gas turbofan engine. In FIG. **2**, a close-up or magnified view of surface **24** of airfoil **18** is shown. Such a close-up or magnified view depicts micro defects in surface **24** of airfoil **18**. If not remedied, such micro defects (e.g., hairline scratches, surface cracks, etc.) can develop in time and lead to catastrophic damage to airfoil **18**. Such remedies can include polishing surface **24**, filling and polishing the damaged area(s), and or replacing airfoil **18**. Such micro defects can be hard to see with the unaided eye. Moreover, inspection of a set of airfoils **18** can be time consuming if done by hand. Inspection of a set of airfoils **18** can be expeditiously performed, however, if done by a plurality of autonomous devices **20**. Such expedience can result from simultaneous inspection of multiple airfoils **18**.

[0017] To perform such an inspection of the surface topology of airfoil **18**, these autonomous devices **20** should be sized appropriately-their lateral dimensions should be less that lateral dimensions of airfoil **18** at inspection locations, and their height (above airfoil **18**) should be less than the space available between adjacent airfoils **18** at such inspection locations. These small autonomous devices **20** can then be deployed as a swarm, so as to quickly and autonomously inspect a plurality of airfoils **18**. Their construction should also be light, consisting of relatively soft materials so that if lost, autonomous device **20** would pass through and disintegrate in the engine while it is in operation causing no damage to the engine. These small autonomous devices **20** should also be light so as to be able to maintain coupling with irregularly shaped airfoils **18**. Without adequate coupling, such autonomous devices **20** could fall from airfoils **18** before inspection has been completed. Moreover, such falls could damage sensitive components of these small autonomous devices **20**. Because these autonomous devices **20** are small, they can be equipped with only limited computing power. Thus, the algorithms used to inspect and/or map the surface topology should be such that they can be performed in-situ in real time using the computing power available to such small autonomous devices **20**. The airfoil contacting members can be designed specifically to simplify the computational requirements for performing image analytics.

[0018] FIGS. **3**A and **3**B are a perspective view and a block diagram, respectively, of an embodiment of an autonomous device configured to inspect the surface topology of an airfoil of a gas turning engine. In FIG. **3**A, autonomous device **20** is depicted from an underside (i.e., a side that will face and engage a surface of an airfoil). Autonomous device **20** includes body or base **26**, means **28** for coupling the body to an airfoil in a manner that permits movement of body **26** across an airfoil, such as airfoil **18** depicted in FIGS. **1** and **2**, while remaining coupled thereto. Autonomous device **20** also includes means **30** for moving body **26** across airfoil **18** while remaining coupled thereto. Autonomous device **20** also includes a plurality of topological sensors **32** extending from body **26** to airfoil **18** when body **26** is coupled thereto. Autonomous device **20** also includes a surface topology calculator **34** (on a topside or bottom side of autonomous device **20**). In the depicted embodiment, body **26** is a substantially planar base to which the various components **28**, **30**, **32** and **34** are all connected.

[0019] In the depicted embodiment, means **28** for coupling body **26** to airfoil **18** includes sealing element **36** and a vacuum pump **38**. Sealing element **36** circumscribes evacuation chamber **39**. Sealing element **36** is configured to engage the surface of the airfoil so as to facilitate evacuation of evacuation chamber **39**. Vacuum pump **38** is configured to evacuate evacuation chamber **39**, thereby causing the autonomous device to adhere to the surface of an airfoil. Sealing element **36** is also configured to slidably move across the airfoil while maintaining vacuum in evacuation chamber **38** so as to adhere autonomous device **20** to the airfoil during movements thereacross. Although two suction elements **36** are depicted in the FIG. **3**A embodiment, a single evacuation chamber **39** provides means **28** for coupling body **26** to a surface of airfoil **18**, two or more such evacuation chambers can be used for such function. For example, in some embodiments, a two or more distinct suction elements **36** can be located on base **18** so as to provide means **28** for coupling base **26** to airfoil **18**.

[0020] Various other means **28** for coupling body **26** to airfoil **18** can include a clip or a band that contacts both body **26** and a backside surface of airfoil **18**, thereby providing a coupling force therebetween. In other embodiments, means **28** for coupling body **26** to airfoil **18** can include contacting members that use a surface force, and/or adhesion to provide coupling between body **26** and airfoil **18**. For example, autonomous device **20** can have such adhering contacting surfaces at ends of legs **22** of autonomous device **20** of the embodiment depicted in FIG. **1**. In such embodiments, contacting surfaces can be of a material or have coating that is tacky when in contact with surfaces **24** of airfoils **18**. In another embodiment, contacting surfaces can provide a van der Waals attractive force with the surface of airfoils **18**.

[0021] In the depicted embodiment, means **30** for moving autonomous device **20** across airfoil **18** include wheels **40** and motor **42**. Wheels **40** extend from body **26** so as to engage the airfoil when body **26** is coupled thereto. Motor **42** is configured to rotate wheels **40**, thereby moving the body across the airfoil when coupled thereto. In the depicted embodiment, means **32** for steering autonomous device **20** as it moves across the airfoil include steering wheel **44** and steering mechanism **46**. A navigation computer can be configured to control means **32** for steering autonomous device **20** and means **30** for moving autonomous device **20** across airfoil **18** so as to inspect and/or map surface topology of a plurality of portions of airfoil **18**. Various sensors, such as edge detectors **50** can be used to provide location information to navigation computer so as to facilitate navigation of autonomous device **20** across airfoil **18** to which it is coupled. In some embodiments, the navigation computer can include or share with topology calculator **34** a processor programmed for performing such operation(s).

[0022] Autonomous device **20** can have various other means for moving autonomous device **20** across airfoil **18**. For example, autonomous device **20** can include articulated legs, as depicted in the FIG. **1** embodiment. In other embodiments, autonomous device **20** can have wheels as depicted in the FIG. **3**A embodiment. Various other arrangements of wheels and/or legs can be used to move autonomous device **20** across airfoil **18**. In other embodiments, means for moving autonomous device **20** across airfoil **18** can include a soft deformable robot body system.

[0023] Autonomous device **20** also includes surface topology calculator **34**, which is attached to body **26** and electrically connected to the plurality of topological sensors **32**. Surface topology calculator **34** is configured to determine a surface topology of airfoil **18** based on the electrical signals generated by the plurality of topological sensors **32**. Surface topology calculator **34** determines the surface topology of airfoil **18** along each of a plurality of paths across airfoil **18** taken by the plurality of topological sensors **32**.

[0024] Surface topology calculator **34** can be configured to execute software, applications, and/or programs stored in computer-readable memory **35**. Examples of surface topology calculator **34** can include one or more of a processor, a microprocessor, a controller, a digital signal processor (DSP), an application specific integrated circuit (ASIC), a field-programmable gate array (FPGA), a video processor, or other equivalent discrete or integrated logic circuitry. In some embodiments, surface topology calculator **34** can include more than one processing core.

[0025] Computer-readable memory **35** is configured to store information and, in some examples, can be described as a computer-readable storage medium. In some examples, a computer-readable memory **35** can include a non-transitory medium. The term “non-transitory” indicates that the storage medium is not embodied in a carrier wave or a propagated signal. In certain examples, a non-transitory storage medium can store data that can, over time, change (e.g., in RAM or cache). In some examples, computer-readable memory **35** can include a temporary memory. As used herein, a temporary memory refers to a memory having a primary purpose that is not long-term storage. Computer-readable memory **35**, in some examples, can include volatile memory. As used herein, a volatile memory refers to a memory that does not maintain stored contents when power to the computer-readable memory **35** is turned off. Examples of volatile memories can include random-access memories (RAM), dynamic random-access memories (DRAM), static random-access memories (SRAM), and other forms of volatile memories. In some examples, computer-readable memory **35** is used to store program instructions for execution by surface topology calculator **34**. Computer-readable memory **35**, in one example, is used by software or applications running on autonomous device **20** to temporarily store information during program execution.

[0026] Computer-readable memory **35**, in some examples, also includes one or more computer-readable storage media. The memory can be configured to store larger amounts of information than volatile memory. The memory can further be configured for long-term storage of information. In some examples, the memory includes non-volatile storage elements. Examples of such non-volatile storage elements can include, for example, magnetic hard discs, optical discs, floppy discs, flash memories, or forms of electrically programmable memories (EPROM) or electrically erasable and programmable (EEPROM) memories.

[0027] Input/output interface **37** is an input and/or output device and enables autonomous device **20** to receive data from an external source or operator and/or to report the results of inspecting airfoil **18** by autonomous device **20**. For example, input/output interface **37** can be configured to receive inputs (e.g., configuration data) from an operator and/or provide outputs related to surface topology of airfoil **18**. Input/output interface **37** can include one or more of a sound card, a video graphics card, a speaker, a display device (such as a liquid crystal display (LCD), a light emitting diode (LED) display, an organic light emitting diode (OLED) display, etc.), a touchscreen, a keyboard, a mouse, a joystick, or other classification of device for facilitating input and/or output of information in a form understandable to users and/or machines. In some embodiments, input/output interface **37** is a network interface. For example, input/output interface **37** can include a network interface card, such as an Ethernet card, an optical transceiver, a radio frequency transceiver, or any other type of device that can send and receive information. Other examples of such network interfaces can include Bluetooth, 3G, 4G, and Wi-Fi radio computing devices as well as Universal Serial Bus (USB). In other embodiments, input/output interface **37** can be a custom network interface.

[0028] Topological sensor **32** can be operated in two ways: as a static device and as a dynamic device. In static operation, the topological sensor **32** generates a signal indicative corresponding to a relationship between the strain and the charge on the piezoelectric crystal of piezoelectric sensor **52**, known as the strain-charge coupling equation. As the surface topology of the airfoil changes (e.g., going over a crack), strain is induced to a part of piezoelectric sensor **52** that is mechanically coupled thereto. This change in the strain on piezoelectric sensor **52** is translated into a change in the electrical properties of the piezoelectric crystal and thus produces a signal indicative thereof. In the dynamic configuration, the piezoelectric sensor is configured to superimpose, to piezoelectric sensor **52**, a sinusoidal loading (i.e., a dynamic loading) to the loading provided by airfoil contacting member **52**.

[0029] FIG. **4** is a schematic diagram of an embodiment of a topological sensor, which is configured for static operation, used to sense a surface topology of an airfoil. In FIG. **4**, topological sensors **32** includes piezoelectric sensor **52**, airfoil contacting member **54**, spring member **56**, and damping member **58** (or shock absorbing member). Piezoelectric sensor **52** is configured to generate an electrical signal indicative of a loading to the piezoelectric sensor **52**. Airfoil contacting member **54** extends between the piezoelectric sensor and the airfoil when body **26** of autonomous device **20** is coupled thereto, thereby providing changes to the loading to piezoelectric sensor **52** in response to changes in the surface topology of the airfoil as the body moves thereacross. In the depicted embodiment, airfoil contacting member **54** includes a wheel configured to ride upon surface **22** of airfoil **18**. In some embodiments, such a wheel is castered (e.g., having a caster) so as to rotate in response to direction of travel. In other embodiments, non-rotating airfoil contacting members can be used, such as for example, a wire-like member, or a leaf-like member, a pointed tip, etc.

[0030] Spring member **56** and damping member **58** together form a linkage assembly that provides mechanical coupling of topological sensor **32** to airfoil **18**. Spring member **56** is configured to determine changes in loading to piezoelectric sensor **52** that result from changes in surface topology as sensed by airfoil contacting member **54**. Damping member **58** is configured to damping oscillations arising from changing surface topology. As the surface topology of the airfoil forces cause surface contacting member **54** to compress spring member **56**, loading to piezoelectric sensor **52** is increased. Piezoelectric sensor **52** then generates a signal indicative of such increased loading thereto. As the surface topology of the airfoil forces cause surface contacting member **54** to compress spring member **56**, loading to piezoelectric sensor **52** is increased. Piezoelectric sensor **52** then generates a signal indicative of such increased loading thereto.

[0031] FIG. **5** is a schematic diagram of an embodiment of a topological sensor, which is configured for dynamic operation, used to sense a surface topology of an airfoil. In FIG. **5**, topological sensor **32** includes piezoelectric sensor **52**, airfoil contacting member **54**, spring member **56**, electrical insulator **60** and piezoelectric actuator **62**. As in the FIG. **4** embodiment, piezoelectric sensor **52** is configured to generate an electrical signal indicative of a loading to the piezoelectric sensor **52**. Airfoil contacting member **54** extends between the piezoelectric sensor and the airfoil when body **26** of the autonomous device **20** is coupled thereto, thereby providing changes to the loading to piezoelectric sensor **52** in response to changes in the surface topology of the airfoil as body **26** moves thereacross. Spring member **56** is configured to determine changes in loading to piezoelectric sensor **52** that result from changes in surface topology as sensed by airfoil contacting member **54**. As the surface topology of the airfoil forces cause airfoil contacting member **54** to compress spring member **56**, loading to piezoelectric sensor **52** is increased. And as the surface topology of the airfoil forces cause surface contacting member **54** to compress spring member **56**, loading to piezoelectric sensor **52** is increased.

[0032] In the FIG. **5** embodiment, topological sensor **32** can be operated in two ways: as a static device and as a dynamic device. In static operation, the signal is provided by a relationship between the strain and the charge on the piezoelectric crystal, known as the strain-charge coupling equation. As the part of piezoelectric sensor **52** that is in contact with the surface changes topology (e.g., going over a crack), the change in the strain on the sensor is translated into change in the electrical properties of the piezoelectric crystal and thus produce a signal. In the dynamic configuration, the piezoelectric sensor is configured to superimpose, to piezoelectric sensor **52**, a sinusoidal loading to the loading provided by airfoil contacting member **52**. To do so, piezoelectric actuator receives a sinusoidal electrical signal and generates a sinusoidal displacement in response to the received sinusoidal electrical signal. This is typically activated at the resonance of the piezoelectric crystal (i.e., where it has a minimum impedance) for maximum sensitivity. The sinusoidal displacement is superimposed upon and displacement of airfoil contacting member **52** caused by changes in the surface topology of the airfoil as autonomous device **20** moves thereacross. Because topological sensor **32** has two stacked piezoelectric crystals—piezoelectric sensor **52** and piezoelectric actuator **62**—and only one is forced to vibrate (i.e., piezoelectric actuator **62**), the electric signal generated by the passive piezoelectric sensor **52** is indicative of both loadings—that induced by piezoelectric actuator **62**, which is modified by the spring mass system. For example, the electrical signal forcing the piezoelectric sensor **52** can be compared with the electrical signal provided to piezoelectric actuator **62**. Differences between these two electrical signals are indicative of the loading provide to piezoelectric sensor **52** by airfoil contacting member **54**. In various embodiments, various metrics of such deviations can be used to indicate the surface topology of the airfoil. For example, differences in the amplitudes, phases, and/or frequency spectra of these electrical signals can be indicative of the surface topology of the airfoil.

[0033] FIG. **6** is an example of topology data of an airfoil that is obtained by a single robot. In FIG. **6**, graph **64** includes horizontal axis **66**, vertical axis **68**, and topology-position relations **70**A-**70**G. Horizontal axis **66** is indicative of the span-wise location, and vertical axis **68** is indicative of relative height (i.e., the chord-wise direction) of the airfoil. Topology-position relations **70**A-**70**G are measurement data produced by five topological sensors **30**′ as autonomous device **52** moves across an airfoil. Each of topology-position relations **70**A-**70**G are indicative of the surface topology as measured along a path across the airfoil taken by the corresponding one of topological sensors **30**′. Note, that each of topology-position relations **70**G-**70**F have a feature **72**B-**72**F indicative of a defect in the airfoil.

Discussion of Possible Embodiments

[0034] The following are non-exclusive descriptions of possible embodiments of the present invention.

[0035] Some embodiments relate to an autonomous device for inspecting surface topology of an airfoil of a gas turbine engine. The autonomous device includes a body, means for coupling the body to the airfoil in a manner that permits movement of the body across the airfoil while remaining coupled thereto, and means for moving the body across the airfoil while remaining coupled thereto. The autonomous device also includes a plurality of topological sensors extending from the body to the airfoil when the body is coupled thereto. Each of the plurality of topological sensors includes a piezoelectric sensor that generates an electrical signal indicative of a loading to the piezoelectric sensor. Each of the plurality of topological sensors also includes an airfoil contacting member extending between the piezoelectric sensor and the airfoil when the body is coupled thereto, thereby providing changes to the loading to the piezoelectric sensor in response to changes in the surface topology of the airfoil as the body moves thereacross. The autonomous device also includes a surface topology calculator attached to the body and electrically connected to the plurality of piezoelectric sensors and configured to locate defects in the surface topology of the airfoil based on the electrical signals generated by the plurality of piezoelectric sensors.

[0036] The system of the preceding paragraph can optionally include, additionally and/or alternatively, any one or more of the following features, configurations and/or additional components:

[0037] A further embodiment of the foregoing system, wherein the surface topology calculator can determine the surface topology of the airfoil along each of a plurality of paths across the airfoil taken by the plurality of topological sensors.
[0038] A further embodiment of any of the foregoing systems, wherein the means for coupling the body to the airfoil can include a device coupling member that couples the body to the airfoil using van der Waals force.
[0039] A further embodiment of any of the foregoing systems, wherein the means for coupling the body to the airfoil can include: a sealing element circumscribing an evacuation chamber, the sealing element configured to engage the surface of the airfoil; and a vacuum pump configured to evacuate the evacuation chamber, thereby causing the autonomous device to adhere to the surface of the airfoil.
[0040] A further embodiment of any of the foregoing systems, wherein the means for moving the body across the airfoil can include: a wheel extending from the body so as to engage the airfoil when the body is coupled thereto; and a motor configured to rotate the wheel, thereby moving the body across the airfoil when coupled thereto.
[0041] A further embodiment of any of the foregoing systems, wherein the means for moving the body across the airfoil can include an arm or leg that sequentially positions itself so as to engage the airfoil and pushes or pulls the body across the airfoil when coupled thereto.
[0042] A further embodiment of any of the foregoing systems, can further include means for steering the body as it moves across the airfoil.
[0043] A further embodiment of any of the foregoing systems, can further include edge sensors that sense the edge of the airfoil.
[0044] A further embodiment of any of the foregoing systems, can further include a navigation computer configured to control the means for steering the body and the means for moving the body across the airfoil so as to map surface topology of a plurality of portions of the airfoil.
[0045] A further embodiment of any of the foregoing systems, wherein the navigation computer can be configured to cause the body to move across the airfoil in a serpentine fashion.
[0046] A further embodiment of any of the foregoing systems, wherein each of the plurality of topological sensors can further include a linkage assembly mechanically coupling the airfoil contacting member to the piezoelectric sensor. The linkage assembly can have: a spring member; and a damping member.
[0047] A further embodiment of any of the foregoing systems, wherein each of the plurality of airfoil contacting members can have a pointed tip configured to contact the airfoil.
[0048] A further embodiment of any of the foregoing systems, wherein each of the contact arms can have a wheel configured to contact the airfoil, the wheel coupled to the contact arm via a castor.
[0049] A further embodiment of any of the foregoing systems, wherein each of the plurality of topological sensors can include a piezoelectric actuator configured to superimpose, to the piezoelectric sensor, a sinusoidal loading to the loading provided by the airfoil contacting member.
[0050] Some embodiments relate to a method for inspecting surface topology of an airfoil of a gas turbine engine. The method includes adhering a base of an autonomous device to the airfoil in a manner that permits movement of the body across the airfoil while remaining coupled thereto. The method continues by moving the base of the autonomous device across the airfoil while remaining coupled thereto. The method includes extending a plurality of topological sensors from the body to the airfoil when the body is coupled thereto. For each of the plurality of topological sensors, the method includes providing changes to a loading to a piezoelectric sensor in response to changes in the surface topology of the airfoil as the body moves thereacross. For each of the plurality of topological sensors, the method also includes generating, via the piezoelectric sensor, an electrical signal indicative of a loading to the piezoelectric sensor. The method includes locating, via a surface topology calculator, defects in the surface topology of the airfoil based on the electrical

signals generated by the plurality of piezoelectric sensors.
[0051] The method of the preceding paragraph can optionally include, additionally and/or alternatively, any one or more of the following features, configurations and/or additional components:
[0052] A further embodiment of the foregoing method can further include superimposing, via a piezoelectric actuator and to the piezoelectric sensor, a sinusoidal loading upon the loading provided by the airfoil contacting member.
[0053] A further embodiment of any of the foregoing methods can further include determining for each of the plurality of topological sensors, via the surface topology calculator, the surface topology of the airfoil along the path traversed thereby, based on the resonant frequency of topological sensor.
[0054] A further embodiment of any of the foregoing methods, wherein coupling the body to the airfoil can include: circumscribing an evacuation chamber via a sealing element circumscribing, the sealing element configured to engage the surface of the airfoil; and evacuating the evacuation chamber via a vacuum pump, thereby causing the autonomous device to adhere to the surface of the airfoil.
[0055] A further embodiment of any of the foregoing methods, wherein moving the body across the airfoil can include: engaging, via a wheel extending from the body, the airfoil when the body is coupled thereto; and rotating the wheel via a motor, thereby moving the body across the airfoil when coupled thereto.
[0056] A further embodiment of any of the foregoing methods, wherein moving the body across the airfoil can include sequentially positioning an arm or leg so as to engage the airfoil, thereby pushing or pulling the body across the airfoil when coupled thereto.
[0057] The invention disclosed herein is not limited to the implementations described but can be practiced with modification and alteration without departing from the scope of the appended claims. Although other embodiments may include not only the specific combinations of features disclosed above but may also include undertaking only a subset of such features, undertaking a different order of such features, undertaking a different combination of such features, and/or undertaking additional features than those features explicitly listed. The scope of the invention should, therefore, be determined with reference to the appended claims, along with the full scope of equivalents to which such claims are entitled.

Claims

1. An autonomous device for inspecting surface topology of an airfoil of a gas turbine engine, the autonomous device comprising: a body; means for coupling the body to the airfoil in a manner that permits movement of the body across the airfoil while remaining coupled thereto; means for moving the body across the airfoil while remaining coupled thereto; a plurality of topological sensors extending from the body to the airfoil when the body is coupled thereto, each of the plurality of topological sensors including: a piezoelectric sensor that generates an electrical signal indicative of a loading to the piezoelectric sensor; an airfoil contacting member extending between the piezoelectric sensor and the airfoil when the body is coupled thereto, thereby providing changes to the loading to the piezoelectric sensor in response to changes in the surface topology of the airfoil as the body moves thereacross; and a surface topology calculator attached to the body and electrically connected to the plurality of piezoelectric sensors and configured to locate defects in the surface topology of the airfoil based on the electrical signals generated by the plurality of piezoelectric sensors.
2. The autonomous device of claim 1, wherein the surface topology calculator determines the surface topology of the airfoil along each of a plurality of paths across the airfoil taken by the plurality of topological sensors.

3. The autonomous device of claim 1, wherein the means for coupling the body to the airfoil includes: a device coupling member that couples the body to the airfoil using van der Waals force.
4. The autonomous device of claim 1, wherein the means for coupling the body to the airfoil includes: a sealing element circumscribing an evacuation chamber, the sealing element configured to engage the surface of the airfoil; and a vacuum pump configured to evacuate the evacuation chamber, thereby causing the autonomous device to adhere to the surface of the airfoil.
5. The autonomous device of claim 1, wherein the means for moving the body across the airfoil includes: a wheel extending from the body so as to engage the airfoil when the body is coupled thereto; and a motor configured to rotate the wheel, thereby moving the body across the airfoil when coupled thereto.
6. The autonomous device of claim 1, wherein the means for moving the body across the airfoil includes: an arm or leg that sequentially positions itself so as to engage the airfoil and pushes or pulls the body across the airfoil when coupled thereto.
7. The autonomous device of claim 1, further comprising: means for steering the body as it moves across the airfoil.
8. The autonomous device of claim 7, further comprising: edge sensors that sense the edge of the airfoil.
9. The autonomous device of claim 8, further comprising: a navigation computer configured to control the means for steering the body and the means for moving the body across the airfoil so as to map surface topology of a plurality of portions of the airfoil.
10. The autonomous device of claim 9, wherein the navigation computer is configured to cause the body to move across the airfoil in a serpentine fashion.
11. The autonomous device of claim 1, wherein each of the plurality of topological sensors further includes: a linkage assembly mechanically coupling the airfoil contacting member to the piezoelectric sensor, the linkage assembly having: a spring member; and a damping member.
12. The autonomous device of claim 1, wherein each of the plurality of airfoil contacting members has a pointed tip configured to contact the airfoil.
13. The autonomous device of claim 1, wherein each of the contact arms has a wheel configured to contact the airfoil, the wheel coupled to the contact arm via a castor.
14. The autonomous device of claim 1, wherein each of the plurality of topological sensors includes: a piezoelectric actuator configured to superimpose, to the piezoelectric sensor, a sinusoidal loading to the loading provided by the airfoil contacting member.
15. A method for inspecting surface topology of an airfoil of a gas turbine engine, the method comprising: adhering a body of an autonomous device to the airfoil in a manner that permits movement of the body across the airfoil while remaining coupled thereto; moving the body of the autonomous device across the airfoil while remaining coupled thereto; extending a plurality of topological sensors from the body to the airfoil when the body is coupled thereto, each of the plurality of topological sensors: providing changes to a loading to a piezoelectric sensor in response to changes in the surface topology of the airfoil as the body moves thereacross; and generating, via the piezoelectric sensor, an electrical signal indicative of a loading to the piezoelectric sensor; and locating, via a surface topology calculator, defects in the surface topology of the airfoil based on the electrical signals generated by the plurality of piezoelectric sensors.
16. The method of claim 15, further comprising: superimposing, via a piezoelectric actuator and to the piezoelectric sensor, a sinusoidal loading upon the loading provided by the airfoil contacting member.
17. The method of claim 15, further comprising: determining for each of the plurality of topological sensors, via the surface topology calculator, the surface topology of the airfoil along the path traversed thereby, based on the resonant frequency of topological sensor.
18. The method of claim 15, wherein the coupling the body to the airfoil includes: circumscribing an evacuation chamber via a sealing element circumscribing, the sealing element configured to

engage the surface of the airfoil; and evacuating the evacuation chamber via a vacuum pump, thereby causing the autonomous device to adhere to the surface of the airfoil.

19. The method of claim 15, wherein moving the body across the airfoil includes: engaging, via a wheel extending from the body, the airfoil when the body is coupled thereto; and rotating the wheel via a motor, thereby moving the body across the airfoil when coupled thereto.

20. The method of claim 15, wherein the moving the body across the airfoil includes: sequentially positioning an arm or leg so as to engage the airfoil, thereby pushing or pulling the body across the airfoil when coupled thereto.